

Roll No.

TBT-101

B. TECH. (BIOTECHNOLOGY) (FIRST SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

FUNDAMENTALS OF MEDICAL MICROBIOLOGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What is flagella ? Classify the bacteria on the basis of flagella using suitable exams. (CO1)

OR

- (b) What are bacterial capsule ? Comment on its composition and different properties and their significance in bacterial pathogenicity. (CO1)

2. (a) What is the difference between dye and stain ? How we can diagnose a *Clostridium tetani* by its morphology ? (CO4)

OR

- (b) What is selective and differential Media ? How you can distinguish different species of *Staphylococcus* using a MSA ? (CO4)

P. T. O.

(2)

3. (a) Classify the bacteria on the basis of shapes and arrangement. Give suitable examples also. (CO1)

OR

- (b) Differentiate between Gram positive and Gram negative bacterial cell wall. Give *two* examples of each Gram positive cocci and Gram negative cocci shaped bacteria. (CO1)
4. (a) Comment on the natural microflora of skin. Also, discuss the significant properties of skin which protects humans against pathogenic microbial infection. (CO1)

OR

- (b) Comment on the routes of entry and modes of transmission of pathogens in human body. (CO1)
5. (a) Discuss the mechanism of classifying *Streptococci* species based on their hemolytic properties using Blood agar medium. (CO1)

OR

- (b) Write the morphology, and clinical manifestation of disease by *Bacillus cerus*. (CO1)

Roll No.

EEC-101

B. TECH. (BIOTECH.) (FIRST SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

BASIC ELECTRICAL AND ELECTRONICS ENGINEERING

Time : 1½ Hours

Maximum Marks : 50

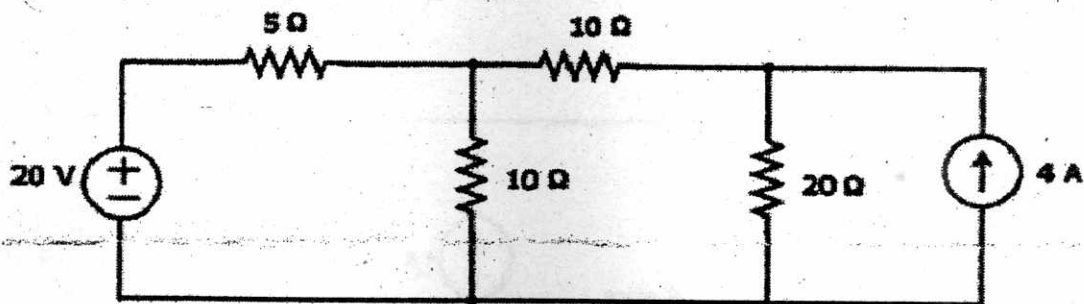
Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain with the formation of energy band diagram in case of metals, insulators and semi-conductors. (CO1)

OR

- (b) State the Superposition Theorem. Calculate current through 20Ω resistor using Superposition theorem in the circuit shown in the figure below : (CO1)



P. T. O.

2. (a) What are star and delta configurations ? Derive expression for star to delta transformation. (CO1)

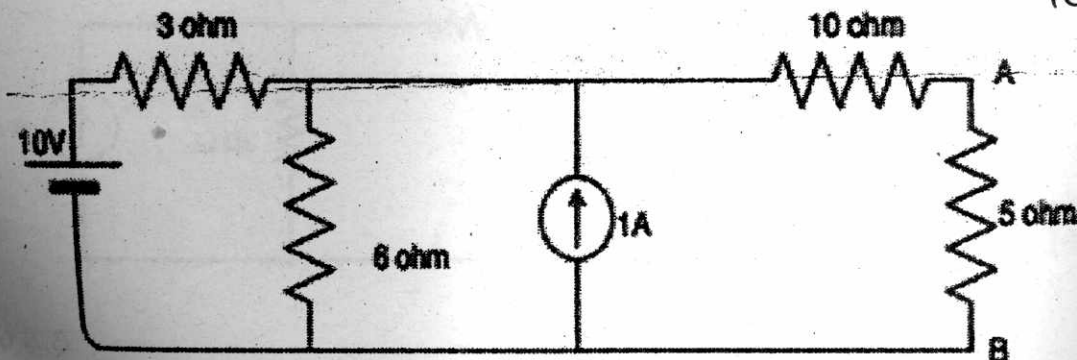
OR

- (b) With the help of a suitable diagram, explain forward biasing condition of a P-N junction diode. (CO1)
3. (a) A donor type impurity concentration of $4.4 \times 10^{20} / \text{m}^3$ is added to a pure Ge semiconductor material. If the intrinsic concentration at 300 K is $2.5 \times 10^{19} / \text{m}^3$ and mobility of electron is $0.8 \text{m}^2 / \text{V-s}$ and mobility of holes is $0.38 \text{m}^2 / \text{V-s}$. Then determine : (CO2)
- (i) Intrinsic conductivity of semiconductor (i.e., conductivity of pure semiconductor before doping)
- (ii) Concentration of majority and minority carriers after doping.

OR

- (b) Write down the statement of maximum power transfer theorem. Derive the condition for maximum power transfer from source to load in DC circuit. Also derive the expression of maximum efficiency in maximum power transfer condition in d.c. circuit. (CO2)
4. (a) State the Norton's theorem. Find out current in 5Ω resistor with the help of Norton's Theorem in the circuit shown in the figure below :

(CO2)



(3)

OR

- (b) Discuss, with the help of a suitable diagram the working of a half wave rectifier. (CO2)
5. (a) Define the following terms along with suitable examples : (CO2)
- (i) Unilateral and Bilateral Elements
 - (ii) Active and Passive Elements
 - (iii) Linear and Non-Linear elements
 - (iv) KCL and KVL

OR

- (b) Define ripple factor. Derive an expression for the average DC value of a load current in case of a half wave rectifier. (CO2)